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#include <windows.h>
#include <GL/glut.h>

char title[] = "3D Shapes";

void initGL() {
glClearColor(0.0f, 0.0f, 0.0f, 1.0f);
glClearDepth(1.0f);
glEnable(GL_DEPTH_TEST);
glDepthFunc(GL_LEQUAL);
glShadeModel(GL_SMOOTH);
glHint(GL_PERSPECTIVE_CORRECTION_HINT, GL_NICEST);
}

void display() {
    glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);

    glMatrixMode(GL_MODELVIEW);

    glLoadIdentity();
    glTranslatef(1.5f, 0.0f, -7.0f);

    glBegin(GL_QUADS);

        glColor3f(0.0f, 1.0f, 0.0f);
        glVertex3f( 1.0f, 1.0f, -1.0f);
        glVertex3f(-1.0f, 1.0f, -1.0f);
        glVertex3f(-1.0f, 1.0f,  1.0f);
        glVertex3f( 1.0f, 1.0f,  1.0f);

        glColor3f(1.0f, 0.5f, 0.0f);
        glVertex3f( 1.0f, -1.0f,  1.0f);
        glVertex3f(-1.0f, -1.0f,  1.0f);
        glVertex3f(-1.0f, -1.0f, -1.0f);
        glVertex3f( 1.0f, -1.0f, -1.0f);

        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex3f( 1.0f,  1.0f, 1.0f);
        glVertex3f(-1.0f,  1.0f, 1.0f);
        glVertex3f(-1.0f, -1.0f, 1.0f);
        glVertex3f( 1.0f, -1.0f, 1.0f);

        glColor3f(1.0f, 1.0f, 0.0f);
        glVertex3f( 1.0f, -1.0f, -1.0f);
        glVertex3f(-1.0f, -1.0f, -1.0f);
        glVertex3f(-1.0f,  1.0f, -1.0f);
        glVertex3f( 1.0f,  1.0f, -1.0f);

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        glColor3f(0.0f, 0.0f, 1.0f);
        glVertex3f(-1.0f, 1.0f, 1.0f);
        glVertex3f(-1.0f, 1.0f, -1.0f);
        glVertex3f(-1.0f, -1.0f, -1.0f);
        glVertex3f(-1.0f, -1.0f, 1.0f);

        glColor3f(1.0f, 0.0f, 1.0f);
        glVertex3f(1.0f, 1.0f, -1.0f);
        glVertex3f(1.0f, 1.0f, 1.0f);
        glVertex3f(1.0f, -1.0f, 1.0f);
        glVertex3f(1.0f, -1.0f, -1.0f);
    glEnd();

    glLoadIdentity();
    glTranslatef(-1.5f, 0.0f, -6.0f);

    glBegin(GL_TRIANGLES);
        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex3f( 0.0f, 1.0f, 0.0f);
        glColor3f(0.0f, 1.0f, 0.0f);
        glVertex3f(-1.0f, -1.0f, 1.0f);
        glColor3f(0.0f, 0.0f, 1.0f);
        glVertex3f(1.0f, -1.0f, 1.0f);

        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex3f(0.0f, 1.0f, 0.0f);
        glColor3f(0.0f, 0.0f, 1.0f);
        glVertex3f(1.0f, -1.0f, 1.0f);
        glColor3f(0.0f, 1.0f, 0.0f);
        glVertex3f(1.0f, -1.0f, -1.0f);

        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex3f(0.0f, 1.0f, 0.0f);
    glEnd();
    glutSwapBuffers();
}

void reshape(GLsizei width, GLsizei height) {

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```
if (height == 0) height = 1;
GLfloat aspect = (GLfloat)width / (GLfloat)height;

glViewport(0, 0, width, height);

glMatrixMode(GL_PROJECTION);
glLoadIdentity();

gluPerspective(45.0f, aspect, 0.1f, 100.0f);
}

int main(int argc, char** argv) {
glutInit(&argc, argv);
glutInitDisplayMode(GLUT_DOUBLE);
glutInitWindowSize(640, 480);
glutInitWindowPosition(50, 50);
glutCreateWindow(title);
glutDisplayFunc(display);
glutReshapeFunc(reshape);
initGL();

glutMainLoop();
return 0;
}
```